



THE CITY APP

Product Requirements Document

AI-Powered Transit, Delivery & Safety Platform for Redemption City, RCCG.
A unified digital coordination layer for intra-city transport, last-mile food delivery,
and GPS-linked emergency response.

PREPARED BY

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VERSION	DATE	AUTHOR	SUMMARY
1.0	June 2026	GodlySage Builders	Initial PRD — drawn from Stage 1 Problem Validation and product design sessions.
1.1	June 2026	GodlySage Builders	Corrected population figure (30,000 to 200,000 inhabitants per verified Stage 1 data), removed unverified location references, updated persona to reflect confirmed estate names.

CORRECTION NOTICE

Version 1.1 corrects three categories of inaccuracy identified against the verified Stage 1 Problem Validation document: (1) the population figure for Redemption City, (2) an unverified zone reference in Persona 1, and (3) a fabricated landmark in Persona 1's pain point. All corrections are sourced from the verified Stage 1 submission which cites the 2023 estimated population of 200,000 inhabitants and confirmed Google Maps estate data. No other factual content has been altered.

SECTION 01

Product Overview & Vision

1.1 Vision Statement

"To become the digital operating system of every structured community in Nigeria, starting with Redemption City — turning fragmented, informal services into a reliable, intelligent, and connected urban experience."

1.2 Product Summary

The City App is an AI-powered, GPS-enabled web platform built for the transportation, food delivery, and emergency safety needs of Redemption City — the RCCG flagship privately-owned city in Ogun State, Nigeria. The city spans approximately 2,500 hectares and is estimated to have approximately 200,000 inhabitants as of 2023, distributed across multiple confirmed residential estates including Haggai Estates 1 through 11, Goshen Estate, Tree of Life Estate, Covenant Estate, and Green Pastures Estate, yet lacks the digital infrastructure to match its size.

Correction (v1.1): Previous version stated "over 30,000 registered residents." The verified Stage 1 Problem Validation document, sourced from Wikipedia citing the RCCG Certificate of Occupancy and the 2023 population estimate, confirms approximately 200,000 inhabitants. The figure 30,000 referred to an earlier 2018 estimate and is not current.

Three problems are addressed through one unified product: the absence of reliable on-demand transport via Keke NAPEP tricycles — especially for residents of peripheral estates across the 2,500-hectare city — no digital food delivery from local vendors, and no real-time GPS-linked emergency alert system. These issues are not isolated; they all stem from the same gap: no shared digital

coordination layer for services within the city.

1.3 Strategic Context

Redemption City is not a hypothetical market. It is a real, growing, high-density settlement with an existing captive population and institutional trust through the RCCG network. The city contains confirmed daily-life infrastructure — Zenith Bank, GTBank, Ecobank, UBA, Redeemer's Health Village Hospital (24-hour), RCCG City Kitchen (Macedonia Road), Emmanuel Park, Redeemer's University — all requiring internal transport to access across 2,500 hectares.

It is both the pilot and the proof of concept. The broader opportunity is a white-label city-mobility platform that replicates across gated communities, university campuses, and estate settlements across Nigeria. Redemption City is the first win. The playbook built here is what scales.

1.4 Platform Type

The City App launches as a mobile-first Progressive Web App (PWA) — accessible via browser without app store installation. This eliminates onboarding friction and works across all smartphone types in the city. A native iOS/Android app is planned for a later phase.

Problem Statement

2.1 The Transportation Gap

Tricycles (Keke NAPEP) are the primary intra-city transport mode in Redemption City. Residents depend on them for everyday commuting — getting to work at the Secretariat, visiting the hospital on RHV Road, reaching the banks (Zenith, GTBank, Ecobank, UBA), accessing restaurants and shops, and navigating between estates including Haggai Estates, Goshen Estate, Tree of Life Estate, and Covenant Estate. However, the current Keke system operates entirely informally, with no digital infrastructure of any kind.

There is no way to summon a driver, no visibility into availability, and no routing logic for peak hours. Operators naturally gravitate only toward the highest-traffic areas of the city, leaving peripheral areas underserved and creating overcrowding, fare disputes, and missed commitments. A verified Google Maps resident review of Haggai Estate 10 states: "The location is far from the main auditorium. You need to be mobile in camp." — confirming the transport gap from real resident experience.

2.2 The Food Delivery Gap

Multiple restaurants, snack shops, and food vendors operate within the city — including RCCG City Kitchen on Macedonia Road and Redemption Resort Dining — but none have a digital delivery channel. Residents cannot order meals to their homes from their favourite city restaurants. Vendors are limited to walk-in customers. The Keke network — already operating city-wide — is an untapped last-mile delivery asset that a digital platform can activate.

2.3 Emergency Response Gap

No unified digital emergency dispatch system exists. In a medical or security emergency, residents rely on personal phone numbers informally shared within the community. There is no GPS-linked alert, no live location sharing with responders, and no structured routing to the right response team. Redemption City has two confirmed medical facilities — Redeemer's Health Village Hospital (24-hour, on RHV Road) and RCCG Health Centre — along with estate security infrastructure, but no digital channel connects residents to these services with a live GPS location.

For an estimated 200,000 inhabitants across 2,500 hectares, this is a critical safety gap.

Correction (v1.1): Previous version stated "a community of over 30,000 residents." Corrected to "an estimated 200,000 inhabitants" per verified Stage 1 data sourced from the 2023 population estimate cited in Wikipedia.

2.4 Root Cause

All three problems share one root: the absence of a shared digital coordination layer for intra-city services. The City App is that layer.

Goals & Objectives

3.1 Product Goals

- **Reliable Mobility** — Reduce average Keke wait time to under 8 minutes via GPS-based real-time dispatch.
- **Commerce Enablement** — Give city food vendors a digital sales and delivery channel beyond walk-in traffic.
- **Life-Safety Infrastructure** — Give every registered resident a one-tap GPS-linked emergency alert connected to city health and security teams.

3.2 Phase 1 Targets (Months 0 to 3)

Objective	Target	How We Measure
Driver onboarding	50+ verified Keke operators	Driver accounts created
Resident adoption	200+ monthly active users	MAU via session tracking
Trip volume	2,000 trips by Month 3	Completed trip events
Wait time	Under 8 minutes	Request-to-pickup delta
Emergency system	Functional at launch	Alerts sent and received in test

3.3 12-Month Strategic Objectives

- Establish The City App as the default transport coordination tool for Redemption City.
- Launch food delivery and onboard 20+ vendors by Month 6.
- Integrate Paystack or Flutterwave wallet for fully cashless transactions.
- Reach platform profitability at the city level (unit economics positive per trip).
- Document a replication playbook for expansion to other RCCG estates and gated communities.

Target Users & Personas

4.1 User Roles

Role	Who They Are	Primary Goal
Users	Registered Redemption City residents and visitors	Book rides, order food, trigger emergency alerts
Transport Provider	Licensed Keke operators registered within the city	Accept ride and delivery requests; earn digitally
Food Vendor	Restaurants, snack shops, and food sellers in the city	List products, receive orders, track sales
Emergency Personnel	City clinic staff and RCCG security unit	Receive and respond to live emergency alerts

4.2 User Personas



Persona 1 — Adaeze, The Working Resident

34, civil servant commuting within the city daily

Age / Role	34, civil servant working at the RCCG National Secretariat
Location	Haggai Estate 10 — confirmed northern section of Redemption City
Pain	Walks a significant distance from her estate to reach a Keke each morning. No certainty one will be available when she arrives at the pickup point. Misses commitments due to unpredictable transit.
Goal	Request a Keke from home and know it is coming before she steps outside
Tech comfort	Uses WhatsApp and OPay daily. Comfortable with mobile web apps.
Done when	Keke arrives in under 8 minutes, booked from inside her house.

Correction (v1.1): Previous version stated location as "Zone 7, northern residential block" — no zone numbering system is confirmed for Redemption City. Corrected to "Haggai Estate 10" which is verified on Google Maps as a confirmed peripheral estate in the northern section of the city. Previous pain point referenced "Car park B" — no such named landmark is confirmed within the city. Removed. Pain point now reflects the verified mobility challenge: significant walking distance from peripheral estates to reach informal Keke pickup points.



Persona 2 — Emeka, The Keke Operator

28, full-time Keke NAPEP operator within the city

Age / Role	28, full-time Keke NAPEP operator within Redemption City
Current behaviour	Parks at high-traffic areas within the city and waits for walk-up passengers
Pain	Unpredictable income, high idle time, no demand visibility across the 2,500-hectare city
Goal	See where passengers are before moving; receive digital payments; complete more trips per day
Tech comfort	Basic Android smartphone. Uses data daily. Needs a fast, simple interface.
Done when	Receives trip requests on his dashboard; earns more per day than before.



Persona 3 — Sister Ngozi, The Food Vendor

45, food vendor operating within the city

Age / Role	45, runs a popular food kitchen within Redemption City
Current behaviour	Serves walk-in customers only; closes early when foot traffic is low
Pain	Loses revenue when residents cannot travel to buy. No digital presence.
Goal	Receive orders digitally, assign Keke for delivery, grow beyond immediate foot-traffic

	proximity
Tech comfort	Uses Facebook and WhatsApp for promotion. Needs a simple vendor dashboard.
Done when	Receiving 5+ delivery orders daily; revenue up within 60 days of listing.



Persona 4 — Officer Dare, City Security Personnel

38, RCCG City Security Officer on shift rotation

Age / Role	38, RCCG City Security Officer on shift rotation
Current behaviour	Receives reports via phone calls and walk-ins only
Pain	No GPS location from callers. Slow response in peripheral estates.
Goal	Receive real-time security alerts with live GPS and resident contact details
Done when	Alert received within 30 seconds. Location visible on map. Contact details immediately accessible.

User Stories

5.1 RESIDENT — TRANSPORT

As a resident, I want to request a Keke from my location so a driver comes to me without me walking to a pickup point.

As a resident, I want to see available drivers on a live map and an estimated fare before confirming a ride.

As a resident, I want to track my driver in real time after booking so I know when they will arrive.

As a resident, I want to rate my driver after a trip.

5.1 RESIDENT — FOOD ORDERING

As a resident, I want to browse vendor menus and place a food order for delivery to my address.

As a resident, I want to track my food order in real time.

As a resident, I want to pay digitally from my wallet so I do not need cash on hand. (Wallet is a placeholder in MVP; live payment processing will be added post-MVP.)

5.1 RESIDENT — EMERGENCY

As a resident, I want a one-tap emergency button on every screen so I can call for help instantly.

As a resident, I want my GPS location and health profile sent automatically with my alert so responders have context before calling me.

As a resident, I want to choose between Medical or Security so my alert reaches the right team.

5.2 TRANSPORT PROVIDER

As a Keke operator, I want to see incoming ride and delivery requests on a dashboard so I can accept jobs efficiently.

As a Keke operator, I want GPS navigation to the pickup location so I do not need to ask for directions.

As a Keke operator, I want to see my daily earnings and trip history.

As a Keke operator, I want to receive ratings from passengers so I maintain good standing.

5.3 VENDOR

As a vendor, I want to list my menu with photos and prices so residents can order from me.

As a vendor, I want to receive orders in real time with items and delivery addresses.

As a vendor, I want to see today's sales and order history on a dashboard.

As a vendor, I want to assign a Keke operator to fulfil a delivery from within the app.

5.4 EMERGENCY PERSONNEL

As health personnel, I want instant notification of health alerts with live GPS and the user's health profile (blood type, allergies, emergency contact) so I can prepare and dispatch.

As security personnel, I want instant notification of security alerts with live GPS so I can dispatch to the incident.

As emergency personnel, I want health and security alerts routed separately so I only see alerts relevant to my role.

Information Architecture

6.1 Platform Structure

Portal	Screens & Modules
Public / Entry	Landing Page, Role Selection, Registration (per role), Sign In, OTP Verification, Password Reset
Resident Portal	Dashboard, Ride Request, Live Map, Provider Matched, Trip in Progress, Post-Trip, Food Browse, Menu, Cart, Checkout, Order Tracking, Emergency Flow (3 steps), Wallet, Favourites, Settings
Transport Portal	Driver Dashboard, Incoming Requests, In-Progress Navigation, Trip History, Earnings, Ratings, Settings
Vendor Portal	Vendor Dashboard, Product Catalogue, Order Management, Sales Summary, Ratings, Storefront Setup, Settings
Emergency Portal	Alert Feed, Alert Detail View, User Directory, Health Info Panel, Broadcast (Admin)

6.2 Navigation Model

Each portal uses a persistent bottom navigation bar on mobile, or a left sidebar on tablet and desktop. The Emergency portal is credential-gated and does not share navigation with any public-facing flow.

The emergency SOS button is pinned as a floating action on every screen in the Resident portal. It is visually distinct — red, labelled SOS — and requires two steps to confirm, preventing accidental triggers.

User Flows

7.1 Resident Onboarding

Land on Landing Page → select Sign Up → Role selection (Resident, Transport Provider, or Vendor) → Registration: full name, phone, email, estate address, blood type, allergies, emergency contact → OTP verification via SMS → Account created → Redirect to Resident Dashboard.

7.2 Ride Request

Open Ride Request from dashboard → Live map loads with resident GPS pinned → Set destination. System shows nearest drivers with ETAs and estimated fare → Confirm request. AI assigns the nearest available driver → Driver accepts. Resident receives driver details and live tracking begins → Ride complete. Fare deducted from wallet (placeholder in MVP). Resident rates driver.

7.3 Food Order

Open Food Order → Browse vendor list by availability or category → Open vendor storefront, browse menu, add to cart → Review cart, confirm delivery address, select payment (wallet — placeholder in MVP) → Place order. Vendor receives notification, confirms, assigns Keke for delivery → Live order tracking → Order delivered → Resident rates vendor.

7.4 Emergency Alert (3-Step)

Resident taps SOS button. Full-screen overlay appears. GPS location captured at this moment — before confirmation → Resident selects alert type: Health or Security → Review screen shows alert type and location. Resident taps Confirm → Alert and GPS coordinates sent to corresponding emergency team dashboard. Team receives push notification within 5 seconds → Personnel opens alert detail: health profile, live map, contact card. Calls resident to confirm. If no answer within 60 seconds, dispatches to live GPS location.

7.5 Driver Dispatch

Driver opens dashboard → Incoming request card shows pickup location, drop-off, and fare → Driver accepts. GPS navigation to pickup begins → Pickup confirmed. Trip in progress → Trip complete. Earnings credited. Driver rates trip.

7.6 Vendor Order Management

Vendor receives real-time notification of incoming order → Opens order detail: items, quantity, delivery address → Confirms and begins preparation. Assigns available Keke operator for delivery → Order dispatched. Status updates pushed to resident → Marked complete on delivery confirmation.

Features

8.1 Module 1 — Dynamic Keke Dispatch (MVP Core)

Feature	Description	Priority
GPS Real-Time Tracking	Live location of all active registered drivers visible on resident and driver maps	P0
Resident Ride Request	Set pickup, set destination, see nearby drivers, confirm request	P0
Driver Dispatch Dashboard	Driver-side view of incoming requests with pickup coordinates and fare	P0
AI Nearest-Driver Matching	Assigns nearest available driver to each incoming request	P0
Live Trip Tracking	Resident tracks driver from acceptance through pickup and dropoff	P0
Digital Fare Calculation	Fare calculated on distance, shown before confirmation	P0
Peak-Period Zone Awareness	Flags high-demand zones at 6–8 AM and 5–7 PM; nudges driver repositioning	P1
Post-Trip Rating	Resident rates driver; affects matching priority	P1
Trip History	Both parties access full trip history with timestamps	P1

8.2 Module 2 — Emergency Alert System (MVP Embedded)

Feature	Description	Priority
Persistent SOS Button	Floating emergency button on every screen of the Resident portal	P0
Two-Step Alert Confirmation	Prevents accidental triggers; requires type selection then final confirm	P0
Automatic GPS Capture	Location captured at first SOS tap, before confirmation	P0
Alert Type Routing	Health to health team; Security to security team. Cross-access blocked.	P0
Health Profile Attachment	Blood type, allergies, emergency contact auto-attached to every alert	P0
Emergency Personnel Portal	Credential-gated portal with alert feed and user directory	P0
Alert Detail View	Live map, health profile, contact card, call button for responders	P0
Non-Response Protocol	Alert flagged unresolved if resident does not answer in 60 seconds; physical dispatch prompted	P1

8.3 Module 3 — Food Delivery (Post-MVP)

Feature	Description	Priority
Vendor Storefront	Profile with logo, hours, location, and menu catalogue	P2
Product Catalogue Management	Add, edit, delete menu items with photo, price, and availability toggle	P2
Resident Browse & Order	Browse vendors by category; add to cart; checkout	P2
Keke Delivery Assignment	Vendor assigns a Keke operator from within the order screen	P2
Live Order Tracking	Resident tracks delivery from dispatch to doorstep	P2
Vendor Sales Dashboard	Daily sales summary, order history, and revenue totals	P2

Functional Requirements

9.1 Authentication & Identity

- **FR-01:** Registration supported for four roles — Resident, Transport Provider, Vendor, Emergency Personnel.
- **FR-02:** Emergency Personnel accounts are provisioned by a system admin and are not reachable via the public sign-up flow.
- **FR-03:** Phone number OTP verification required before any account accesses its dashboard.
- **FR-04:** Password reset must be available via phone number verification.
- **FR-05:** Session tokens expire after 30 days of inactivity.

9.2 GPS & Mapping

- **FR-06:** GPS permission requested and stored at first login for Resident and Driver roles.
- **FR-07:** Live map shows all active drivers within the city geofence, updating every 5 seconds maximum.
- **FR-08:** Ride request flow allows residents to confirm or manually adjust their GPS pickup pin.
- **FR-09:** GPS coordinates captured at the moment the SOS button is tapped, before the user confirms the alert.

9.3 Dispatch & Matching

- **FR-10:** AI matching assigns the nearest available driver within 10 seconds of request confirmation.
- **FR-11:** Declined requests are re-queued to the next nearest driver automatically.
- **FR-12:** A driver cannot receive a new request while an active trip or delivery is in progress.
- **FR-13:** System flags high-demand zones during peak windows (6 to 8 AM, 5 to 7 PM) and surfaces heat indicators on the driver map.

9.4 Emergency System

- **FR-14:** SOS button present and functional on every screen of the Resident portal.
- **FR-15:** Alert type determines routing destination; cross-role access to alert details is blocked.
- **FR-16:** Every alert includes: resident name, phone, GPS coordinates, blood type, allergies, and emergency contact.
- **FR-17:** Emergency Personnel receive a push notification within 5 seconds of alert submission.
- **FR-18:** Emergency dashboard displays alerts in reverse chronological order with status indicators (New, In Progress, Resolved).

9.5 Wallet

- **FR-19:** Residents can view wallet balance and transaction history from the dashboard. (Wallet is a placeholder in MVP; no live transactions in Phase 1.)
- **FR-20:** Fare deduction occurs at trip completion, not at booking.
- **FR-21:** Paystack/Flutterwave integration activates in Phase 2 pending gateway setup and regulatory review.
- **FR-22:** Vendors see revenue per order and cumulative daily totals on their dashboard.

Non-Functional Requirements

Category	Requirement	Target
Performance	Map and dispatch screen load time	Under 3 seconds on 3G
Performance	Driver matching response time	Under 10 seconds from request confirmation
Performance	Emergency alert delivery	Under 5 seconds from submission to personnel dashboard
Reliability	Platform uptime	99.5% monthly (Phase 1)
Reliability	GPS tracking refresh	5-second maximum on active trips
Scalability	Concurrent users at launch	500 minimum
Scalability	Concurrent users at 12 months	5,000+
Security	Data storage	Encrypted at rest (AES-256); HTTPS only
Security	Health profile data	Restricted to Emergency Personnel via RBAC; never visible to drivers or vendors
Security	Emergency portal access	Separate credentials; not accessible via public sign-up
Usability	Onboarding to first action	Under 5 minutes for any role
Usability	SOS tap to alert sent	Under 20 seconds end-to-end
Accessibility	Text contrast	WCAG AA minimum
Platform	Delivery method	Progressive Web App (PWA), mobile-first
Platform	Browser support	Minimum Chrome 90+, Safari 14+, Firefox 88+
Offline	Offline fallback	Emergency contact info and last-known driver state cached offline

Assumptions & Constraints

11.1 Assumptions

- Residents own smartphones with mobile data. The city has sufficient network coverage for GPS and data-dependent features across its 2,500-hectare boundary.
- RCCG city leadership will support Keke operator onboarding and emergency personnel integration.
- Keke operators are willing to register in exchange for higher daily trip volume and digital earnings.
- Food vendors will accept a 5% commission model for delivery reach they cannot achieve independently.
- Emergency personnel have smartphones and can operate the portal after brief onboarding.
- The city boundary is definable as a geofence to constrain dispatch logic to the correct zone.
- Phase 1 wallet is a UI shell only. Real money movement begins in Phase 2 after payment gateway integration.

11.2 Constraints

- **Timeline:** Phase 1 must be prototype-ready for Kingdom Hack 3.0 and production-ready within 3 months post-competition.
- **Team capacity:** Phase 1 scope must be achievable with the current GodlySage team. Phase 2 features depend on funding or team growth.
- **Payment regulation:** Live payment processing requires CBN compliance review. Wallet activation is Phase 2.
- **GPS accuracy:** Low-end Android devices may produce GPS drift up to 15 metres. The matching algorithm must tolerate this.
- **Network:** Data speeds in parts of the city may be inconsistent. The platform must degrade gracefully on slow connections.
- **Data privacy:** Health profile data must never appear on any driver or vendor screen. RBAC enforcement is required at the API level, not just the UI.

Success Metrics & KPIs

12.1 Phase 1 KPIs (Months 0 to 3)

KPI	Baseline	Target	Measurement
Average ride wait time	Walk-to-pickup (untracked)	Under 8 minutes	Request-to-pickup delta
Registered Keke operators	0 (no platform)	50+	Verified driver accounts
Monthly active residents	0 (no platform)	200+	MAU via session tracking
Trips completed	0	2,000 by Month 3	Completed trip events
Emergency system	Non-existent	Live; 0 missed alerts in test	Alert send vs. acknowledge rate
Driver average rating	N/A	3.8+ stars	Post-trip resident ratings

12.2 Phase 2 KPIs (Months 3 to 6)

KPI	Target	Measurement
Onboarded food vendors	20+ active storefronts	Vendor accounts with at least 1 published menu item
Food delivery orders/month	500+ by Month 6	Completed order events
Live wallet transactions	Active post-integration	Paystack/Flutterwave transaction logs
Platform gross revenue	Revenue positive by Month 5	Trip + delivery commissions
Vendor average rating	4.0+ stars	Post-delivery resident ratings

12.3 Guardrail Metrics

- Emergency alert delivery must never exceed 10 seconds at any load level. Exceeding this is a P0 incident.
- Health profile data must never appear on any driver or vendor screen. A confirmed data leak triggers immediate platform review.
- Driver acceptance rate must stay above 70%. Below this signals a supply imbalance requiring active intervention.

MVP Scope

13.1 What is In and What is Out

MVP — IN SCOPE

Resident registration with health profile
Transport provider registration and onboarding
GPS-based ride request (resident side)
Driver dispatch dashboard with incoming requests
AI nearest-driver matching
Live trip tracking (resident and driver)
Emergency SOS button with 3-step confirmation flow
Alert routing to emergency personnel dashboard
Emergency personnel portal with user directory and health info
OTP phone verification for all roles
Wallet UI shell (balance display only — no live transactions)
Post-trip rating system

POST-MVP — EXCLUDED

Live wallet payment processing (Paystack/Flutterwave)
Food delivery module
Vendor storefront and product catalogue
Native iOS/Android apps
Multi-language support (Yoruba, Pidgin)
Vendor payout system
Admin analytics panel
Driver background verification workflow
Scheduled/advance ride booking
In-app messaging between resident and driver
Coupon and promotional code system

13.2 Notes for the Development Team

- **Build GPS first.** Both the dispatch system and the emergency alert system depend on reliable GPS. Establish map rendering, geofence logic, and location permissions in Week 1.
- **Isolate the emergency portal from day one.** Do not share session infrastructure or navigation between the emergency portal and any public-facing portal. Isolation is a safety requirement.
- **Design for low-end Android.** The driver dashboard must be fast and readable on entry-level devices over a 3G connection.
- **Keep the SOS flow under 20 seconds** from tap to alert sent. If it takes longer, residents will not trust it in a real emergency.
- **Build wallet data models in Phase 1** even though the feature is dormant. Retrofitting payment infrastructure later is expensive.

Dependencies & Risks

14.1 Technical Dependencies

- **Mapping and GPS:** OpenStreetMap with Leaflet.js and geofence logic for the Redemption City boundary.
- **SMS OTP:** Africa's Talking, Termii, or Twilio for phone number verification.
- **Push Notifications:** Firebase Cloud Messaging (FCM) for real-time driver, vendor, and emergency alerts.
- **Payment Gateway (Phase 2):** Paystack or Flutterwave — both CBN-licensed Nigerian payment processors.
- **Backend:** FastAPI (Python) with PostgreSQL; WebSockets for real-time GPS and dispatch events.

14.2 External Dependencies

- RCCG city administration approval for emergency personnel integration and official platform endorsement.
- Keke operators association engagement for driver onboarding.
- City clinic and security leadership sign-off on the emergency portal and alert protocols.

14.3 Risk Register

Risk	Likelihood	Impact	Mitigation
Low Keke operator adoption	Medium	High	Partner with operators association leadership; guarantee trip volume; free onboarding.
GPS accuracy on low-end devices	High	Medium	15-metre tolerance in matching algorithm; manual pickup pin correction in UI.
Emergency alert delivery failure	Low	Critical	Redundant push and SMS fallback for all alerts. Never single-channel.
Payment gateway regulatory delay	Medium	Medium	Wallet UI ships Phase 1; live payments go Phase 2. Cash fallback maintained.
Resident data privacy breach	Low	Critical	Health data RBAC at API level, not just UI. Security audits from Phase 1.
Competition from informal WhatsApp groups	Medium	Low	GPS dispatch, ratings, and emergency integration are capabilities WhatsApp cannot replicate.

Glossary

Term	Definition
Keke NAPEP	Three-wheeled motorised tricycle; primary intra-city transport mode in Redemption City.
PWA	Progressive Web App. A web app accessible via browser without an app store install, behaving like a native mobile app.
RCCG	Redeemed Christian Church of God. Parent organisation of Redemption City.
Geofence	A virtual boundary around Redemption City's 2,500-hectare area used to constrain dispatch logic to the correct operational zone.
OTP	One-Time Password. A six-digit code sent via SMS to verify a user's phone number at registration.
AI Matching	The nearest-driver assignment algorithm that evaluates GPS location, availability, and current trip state to assign requests.
SOS Button	The persistent emergency alert button visible on every screen of the Resident portal.
RBAC	Role-Based Access Control. Ensures each user role only accesses data and screens appropriate to their function.
MAU	Monthly Active Users. Unique users completing at least one session in a calendar month.
Paystack / Flutterwave	CBN-licensed Nigerian payment gateways for wallet top-up and vendor payouts (Phase 2).
MVP	Minimum Viable Product. The Phase 1 build covering Keke dispatch, GPS tracking, and the emergency alert system.
Haggai Estate 10	A confirmed peripheral residential estate in the northern section of Redemption City, verified on Google Maps. Used as a representative example of the transport access gap.
RHV Road	Redeemer's Health Village Road. The confirmed address of Redeemer's Health Village Hospital within Redemption City.

END OF DOCUMENT

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